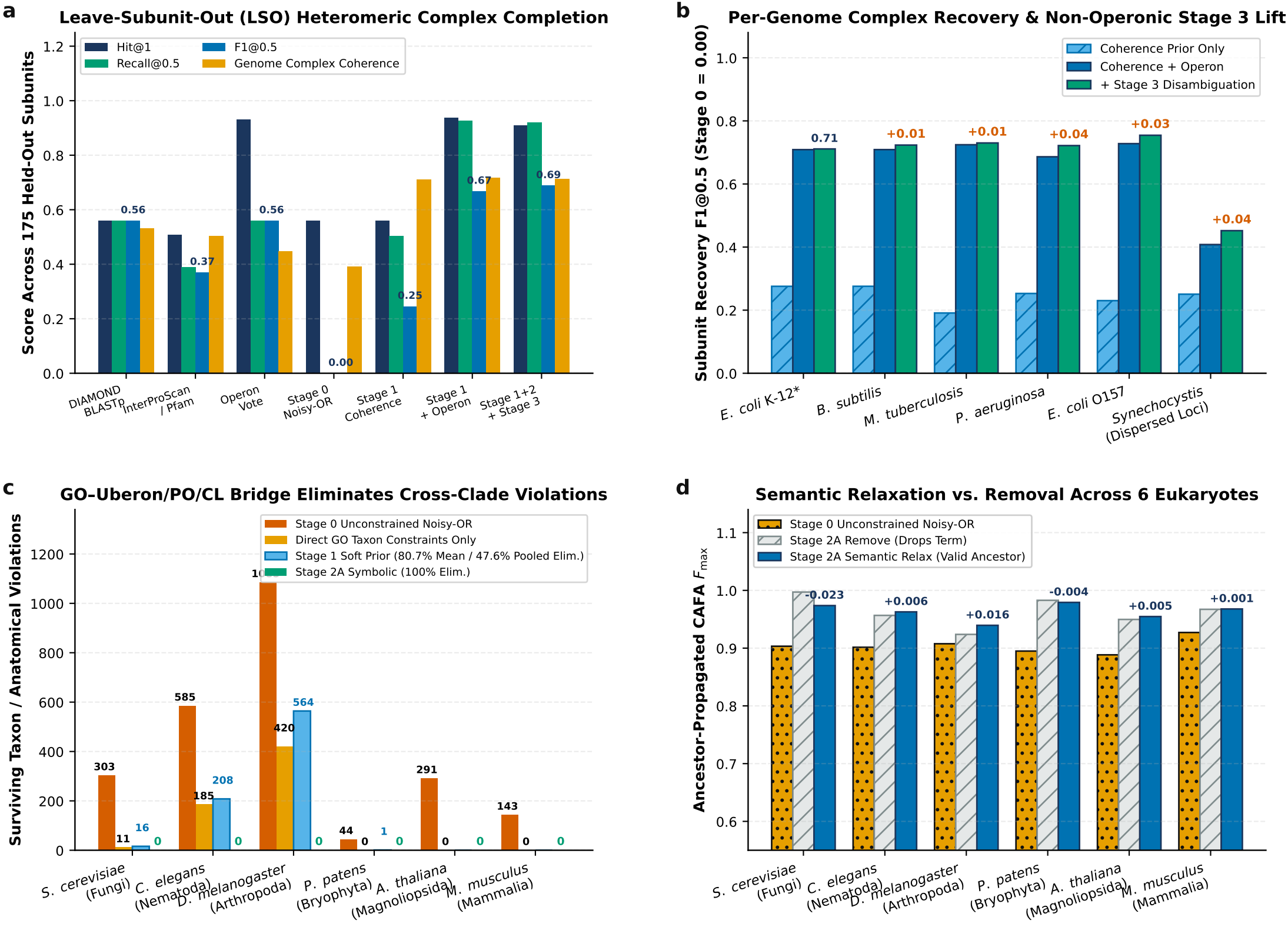

a
Leave-Subunit-Out (LSO) Heteromeric Complex Completion
Score Across 175 Held-Out Subunits
Hit@1
Recall@0.5
F1@0.5
Genome Complex Coherence
0.56
0.37
0.56
0.00
0.25
0.67
0.69
DIAMOND BLASTp
InterProScan / Pfam
Operon Vote
Stage 0 Noisy-OR
Stage 1 Coherence
Stage 1 + Operon
Stage 1+2 + Stage 3
b
Per-Genome Complex Recovery & Non-Operonic Stage 3 Lift
Subunit Recovery F1@0.5 (Stage 0 = 0.00)
Coherence Prior Only
Coherence + Operon
+ Stage 3 Disambiguation
0.71
+0.01
+0.01
+0.04
+0.03
+0.04
E. coli K-12*
B. subtilis
M. tuberculosis
P. aeruginosa
E. coli O157
Synechocystis (Dispersed Loci)
c
GO-Uberon/PO/CL Bridge Eliminates Cross-Clade Violations
Surviving Taxon / Anatomical Violations
Stage 0 Unconstrained Noisy-OR
Direct GO Taxon Constraints Only
Stage 1 Soft Prior (80.7% Mean / 47.6% Pooled Elim.)
Stage 2A Symbolic (100% Elim.)
303
11
16
0
585
185
208
0
420
564
0
44
0
1
0
291
0
0
143
0
0
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)
d
Semantic Relaxation vs. Removal Across 6 Eukaryotes
Ancestor-Propagated CAFA F_{max}
Stage 0 Unconstrained Noisy-OR
Stage 2A Remove (Drops Term)
Stage 2A Semantic Relax (Valid Ancestor)
-0.023
+0.006
+0.016
-0.004
+0.005
+0.001
S. cerevisiae (Fungi)
C. elegans (Nematoda)
D. melanogaster (Arthropoda)
P. patens (Bryophyta)
A. thaliana (Magnoliopsida)
M. musculus (Mammalia)